

A Harmonic-Mixing PLL Architecture for Millimeter-Wave Application

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Abstract—A low-noise synthesizer design in the millimeter-wave (mm-wave) range is complicated by the invariably large closed-loop gain and the high operation frequency of the voltage-controlled oscillator (VCO). To overcome these challenges, this work proposes a harmonic-mixing synthesizer system with a 188-dB figure-of-merit mm-wave VCO. The synthesizer extends the bandwidth to 5 MHz, thereby suppressing VCO noise. Importantly, DSM is not amplified by the $f_{\text{VCO}}/f_{\text{ref}}$ gain of the system. A prototype employing a 74-MHz reference achieves a -250 -dB phase-locked loop (PLL) figure of merit with a measured 88-fs rms jitter.

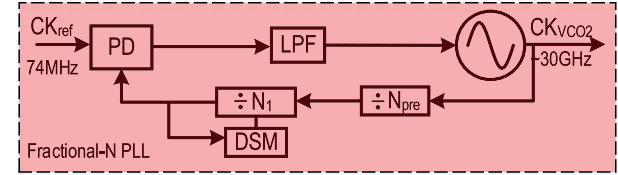
Index Terms—coupled voltage-controlled oscillator (VCO), delta-sigma modulator (DSM), fractional- N PLL, frequency synthesizer, harmonic mixing, heterodyne phase-locked loop (PLL), millimeter wave (mm-wave), PLL noise optimization, subsampling.

I. INTRODUCTION

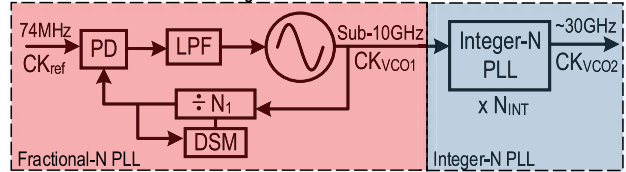
A PHASE-LOCKED loop (PLL) is a feedback system where the dividing ratio, $N = f_{\text{VCO}}/f_{\text{ref}}$, defines the feedback ratio $1/N$ and the closed-loop gain N . In modern radios, this system is used to multiply a sub-100-MHz crystal (XTAL) reference frequency (f_{ref}) up to a target carrier frequency. However, phase noise and spur amplification accompany the frequency multiplication. The large N used to synthesize millimeter-wave (mm-wave) output frequencies will lead to an unacceptably large scaling of phase noise in the reference input and of spurs created by imperfections in the PLL. For instance, suppose that we have available a 6-GHz synthesizer that manifests -60 -dBc fractional spur and -50 -dBc integrated phase noise (IPN). After its output frequency is multiplied to 28 GHz by a hypothetical noiseless PLL, the frequency multiplied tone will exhibit a -46.6 -dBc spur and -36.6 -dBc IPN.

Noise and spurs from the reference input of PLL can, of course, be lowered by narrowing PLL bandwidth; but this is at the cost of increasing the contribution of voltage-controlled

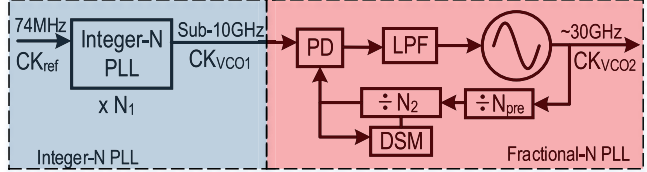
a. Arch1: one stage fractional- N PLL



b. Arch2: fractional- N PLL + integer- N PLL



c. Arch3: integer- N PLL + fractional- N PLL



d. Our Arch: integer- N PLL + harmonic-mixing PLL

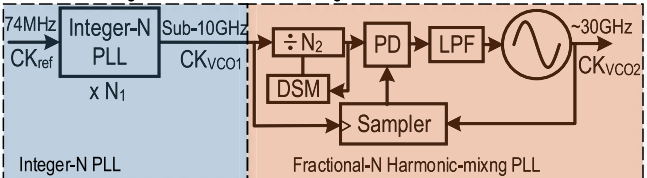


Fig. 1. Possible choice of mm-wave PLL architectures.

oscillator (VCO) noise. Moreover, the oscillator figure of merit (FoM_{VCO}) degrades in the mm-wave range due to the reduced quality factors (Q 's) of switched-capacitor banks and other parasitics [1], [2], [3], [4], [5]. Therefore, after the bandwidth optimization that balances VCO noise and the reference noise, the overall noise performance in a single-stage PLL (Arch1 in Fig. 1) at mm-wave is invariably worse than equivalent sub-10-GHz designs. The bandwidth optimization tradeoff is further complicated when delta-sigma modulator (DSM) noise is introduced, which can experience additional amplification in mm-wave due to the presence of prescalers [6], [7], [8], [9], [10], [11], [12]. Although DSM noise can be canceled with digital circuits [7], [8], [9], [12], [13], it comes at the cost of complexity, additional noise source, and settling time.

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A straightforward way to improve the performance of any PLL is to decrease the closed-loop gain N by increasing the reference frequency [6], [7], [8], [12], [14], [15]. However, the frequency of inexpensive crystal (XTAL) resonators, which are almost always used in commercial radios, is limited to around ~ 100 MHz. Given this, two-stage or multistage designs should be considered for mm-wave applications. In [13], [14], and [16], a first-stage fractional- N PLL is followed by a second-stage integer- N PLL operating at mm-wave frequencies [Fig. 1(b)]. The second stage acts as a simple frequency multiplier and suppresses the noise of the mm-wave VCO with a large bandwidth. However, this large bandwidth also amplifies, without meaningful filtering, the spurs and noise (including DSM noise) of the first stage. Alternatively, as shown in Fig. 1(c), some designs [17], [18] swap the PLLs, cascading a first-stage integer- N PLL with a second-stage fractional- N PLL. Now, the DSM clocks at a much higher frequency, so its quantization noise is accordingly lower. However, there is a limit to the speed of DSM and the multi-modulus divider (MMD) operation, which restricts the output frequency of the first stage and generally necessitates the use of a prescaler. This, in turn, degrades the noise performance of the second stage. Oversampling the reference is another approach to effectively raise the reference frequency [19], [20], [21]. However, it relies heavily on calibration in fractional- N uses.

In this work, to relax the tradeoff between bandwidth and noise amplification, we propose a first-stage integer- N PLL in cascade with a second-stage harmonic-mixing PLL (HMPLL) [22], [23], [24] [Fig. 1(d)]. Uniquely, the HMPLL has close-to-unity closed-loop gain, so DSM noise and fractional- N spurs do not scale up. This essential property extends loop bandwidth to suppress the VCO noise without compromise. The mm-wave prototype being reported here [25] is fabricated in 7-nm CMOS and demonstrates a -250 -dB FoM_{PLL} , 10 dB better than the comparable published numbers.

This article is organized as follows. Section II discusses PLL noise optimization in the context of mm-wave. Section III reviews and elaborates on HMPLL architecture and its application to mm-wave. Section IV deals with the frequency plan, while Section V introduces a coupled mm-wave VCO that has increased immunity to pulling. Sections VI and VII discuss other critical circuit blocks, and Section VIII presents the measurement results.

II. THEORY OF PLL NOISE OPTIMIZATION

As pointed out in [26] and [27], the loop gain of high-order PLLs can be simplified to the 1st order with only one key design parameter, the unity-gain frequency f_u , which also defines the PLL bandwidth f_{BW} . Such a loop high-pass filters the VCO noise while low-pass filtering the phase noise at the loop's reference input.

Based on [28], the IPN of the simplified loop is

$$\langle \phi_{\text{total}}^2 \rangle = \pi \left[N^2 f_{\text{BW}} \text{PN}_{\text{in}} + \frac{10^{-\frac{\text{FoM}_{\text{VCO}}}{10}} f_{\text{VCO}}^2}{\frac{P_{\text{VCO}}}{1 \text{ mW}} f_{\text{BW}}} \right]. \quad (1)$$

PN_{in} is the total input-referred noise, including the noise of reference and loop components such as phase detector (PD),

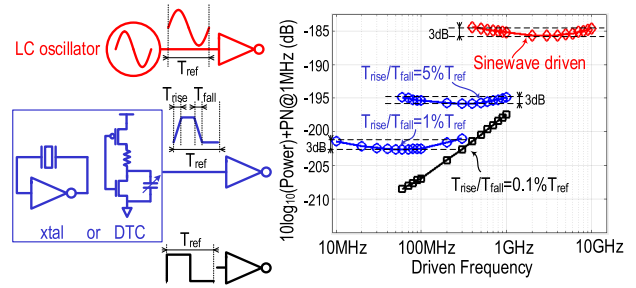


Fig. 2. Effect of f_{ref} transition time to the product of power and phase noise of the cascaded buffer.

divider, and loop filter. FoM_{VCO} is the VCO FoM, which is defined in [28], [29], and [30]. P_{VCO} is the power of VCO in milli-watts. N is the dividing ratio, and f_{VCO} is the output frequency. The IPN is minimized at the optimal bandwidth of

$$f_{\text{BW}_{\text{opt}}} = \sqrt{\frac{10^{-\frac{\text{FoM}_{\text{VCO}}}{10}}}{\frac{P_{\text{VCO}}}{1 \text{ mW}} \text{PN}_{\text{in}}}} \times f_{\text{ref}} \quad (2)$$

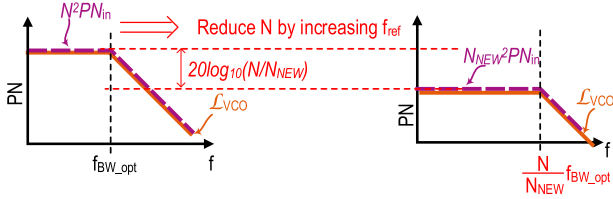
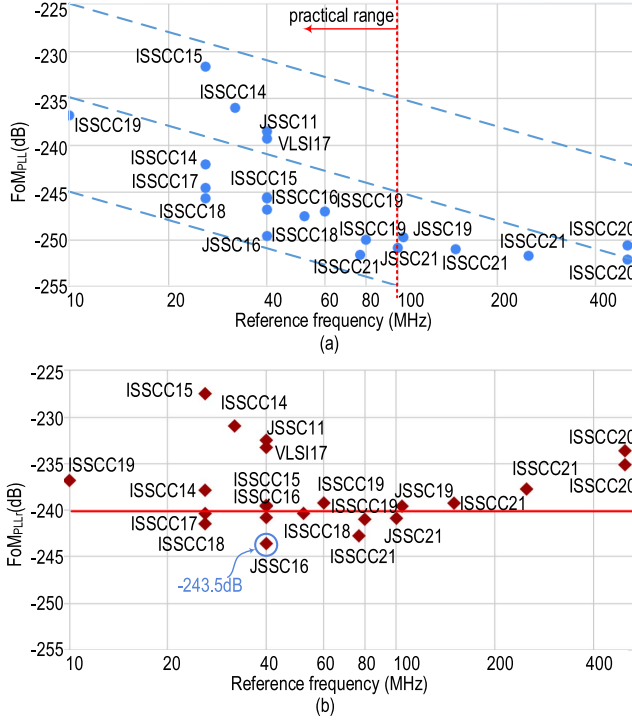
where the spectrum of two noises overlaps, leading to the optimal IPN

$$\langle \phi_{\text{total, opt}}^2 \rangle = \frac{2\pi f_{\text{VCO}}^2}{f_{\text{ref}}} \sqrt{\frac{10^{-\frac{\text{FoM}_{\text{VCO}}}{10}} \text{PN}_{\text{in}}}{P_{\text{VCO}}/1 \text{ mW}}}. \quad (3)$$

Gao et al. [28] have shown that the product of buffer power and its phase noise is proportional to f_{ref}^2 under the assumption that the zero-crossing slope of a clock buffer is independent of the input slew rate. In practice, the input transition time affects the output. As shown in Fig. 2, either driven by an XTAL oscillator or DTC whose rising and falling transition is 1%-5% of one cycle, or when driven by a sinusoidal LC oscillator, the output buffer's phase noise and power product is almost constant without dependence on the clock frequency. When the input transition is less than 0.1% of a cycle, the product tracks f_{ref}^2 . In this case, the buffer will often contribute negligible noise and consume minimal power. Besides, the performance of all the loop components does not track f_{ref}^2 as the charge pump in [28] does. For example, the gated ring oscillator TDC quantization noise [31] is inversely proportional to f_{ref} , while its power is proportional to f_{ref} , leading to a constant product. In the conventional XOR-detector-based PLL, the low-pass filter (LPF) resistance determines power and noise and has no dependency on f_{ref} . Even in a charge pump, when not charging the loop capacitor, the current source is often kept ON for good linearity [32], [33]. Thus, the noise-power product does not follow f_{ref}^2 . Given all those arguments and simulations, it is reasonable to assume that PN_{in} , in general, is not affected by f_{ref} . Therefore, with an overall PLL power of P_{total} , the optimal FoM_{PLL} based on [28] is

$$\text{FoM}_{\text{PLL}} = 10\log_{10} \left[\frac{\sqrt{10^{-\frac{\text{FoM}_{\text{VCO}}}{10}} \text{PN}_{\text{in}} \times P_{\text{total}}}}{2\pi f_{\text{ref}} \sqrt{P_{\text{VCO}} \cdot 1 \text{ mW}}} \right]. \quad (4)$$

$-10\log_{10} f_{\text{ref}}$ in (4) can be understood in Fig. 3. Raising f_{ref} directly lowers the dividing ratio from N to N_{NEW} , which

Fig. 3. f_{ref} 's effect to FoM_{PLL} .Fig. 4. Survey of fractional- N PLL FoM_{PLL} and FoM_{PLLr} versus f_{ref} .

reduces the noise contribution of PN_{in} by $20 \log_{10}(N/N_{\text{NEW}})$. The bandwidth has to be increased to further suppress the VCO noise and reach the new optimum. Thus, the overall area under the phase noise spectrum is only reduced by $10 \log_{10}(N/N_{\text{NEW}})$. As shown in Fig. 4(a), FoM_{PLL} of published designs typically benefit from a higher reference frequency, validating (4). In practice, however, when f_{ref} is taken from commercial XTALs today, it will not exceed 100 MHz. Thus, published PLLs that use higher reference frequencies are for our purposes irrelevant, incomplete, or exploit non-mainstream technologies (film bulk acoustic resonator (FBAR) resonators). In light of this, a better or fairer metric of the absolute performance of any given PLL architecture is one that normalizes also for f_{ref} , i.e.,

$$\text{FoM}_{\text{PLLr}} = \text{FoM}_{\text{PLL}} + 10 \log_{10} \left(\frac{f_{\text{ref}}}{10 \text{ MHz}} \right). \quad (5)$$

When (5) is applied to the same PLLs as shown in Fig. 4(b), the performance is largely independent of f_{ref} , that is, superior performance was largely due to the use of a higher reference frequency during measurement. After eliminating the f_{ref} -dependency, mm-wave PLL performance relies heavily on FoM_{VCO} and PN_{in} , which are not strong functions of

technology. Be that as may, an mm-wave PLL's FoM_{VCO} is always lower than in the sub-10-GHz range; thus, a single-stage PLL architecture will perform poorly. As to lower PN_{in} , an integer- N PLL is the best because it has the fewest noise sources originating in the essential building blocks, i.e., phase/frequency detector (PFD), frequency divider, and loop LPF. Extra circuitry will typically degrade PN_{in} .

DSM quantization noise will cause FoM_{PLLr} to degrade further. When referred to the input of the PLL, the quantization noise is given by [34]

$$\mathcal{L}_{\text{DSM}, \text{in}} = \frac{\pi^2}{3 f_{\text{DSM}} N_{\text{mmd}}^2} \left(2 \sin \left(\frac{\pi f}{f_{\text{DSM}}} \right) \right)^{2(m-1)} \quad (6)$$

where N_{mmd} is the dividing ratio of the MMD, which is toggled by the DSM, and f_{DSM} is the operating frequency of DSM, which is often the reference frequency. Normally, N_{mmd} is equal to $N = f_{\text{VCO}}/f_{\text{ref}}$. In the mm-wave design, the added prescaler causes a smaller N_{mmd} than N . This degrades PN_{in} and the mm-wave PLL performance. DSM noise cancellation can be used to suppress $\mathcal{L}_{\text{DSM}}$. However, this requires use of a DTC [8], [13], [35], TDC [36], or analog DAC [37], all of which introduce extra noise source and worsen PN_{in} .

This backdrop helps place this work in context. It shows the development of a new architecture for an mm-wave frequency synthesizer that uses a practical reference frequency (~ 80 MHz) and improves FoM_{PLLr} by pushing the PN_{in} and FoM_{VCO} close to a sub-10-GHz integer- N PLL.

III. HARMONIC-MIXING PLL

A. Frequency Translation and Phase Detection in Mixing

Mixing two sinusoidal signals (u_1 and u_2) produces new sinewaves at the sum and difference of the original frequencies. An LPF following the mixer selects the difference frequency (7):

$$\begin{aligned} & \underbrace{\cos(\omega_1 t + \theta_1(t))}_{u_1(\omega_1, \theta_1)} \cdot \underbrace{\cos(\omega_2 t + \theta_2(t))}_{u_2(\omega_2, \theta_2)} \\ & \quad \Downarrow \text{LPF selects} \\ & \frac{1}{2} \underbrace{\cos((\omega_1 - \omega_2)t + \theta_1(t) - \theta_2(t))}_{u_3(\omega_{\text{IF}}, \theta_{\text{err}})}. \end{aligned} \quad (7)$$

If ω_1 is not equal to ω_2 , the carrier frequency is translated to IF as

$$\omega_{\text{IF}} = \omega_1 - \omega_2. \quad (8)$$

As indicated in (7), θ_{err} , the difference of θ_1 and θ_2 , phase-modulates the IF carrier. The amplitude of IF, on the other hand, is constant and contains no information. The Laplace-domain (S -domain) signal flow of the phase information from mixer inputs to the output is a simple subtraction, as shown in Fig. 5(a).

When ω_1 equals ω_2 , unlike the previous case, θ_{err} directly creates a voltage at the mixer output. The mixer detects a phase difference as a non-zero voltage [38]. The mixer is modeled as a subtraction block cascaded by a gain block. The PD responds with a static voltage for a constant/dc phase error ($\Phi_{\text{err}}(s=0)$) between the two inputs. The phase detection gain K_{PD} is the

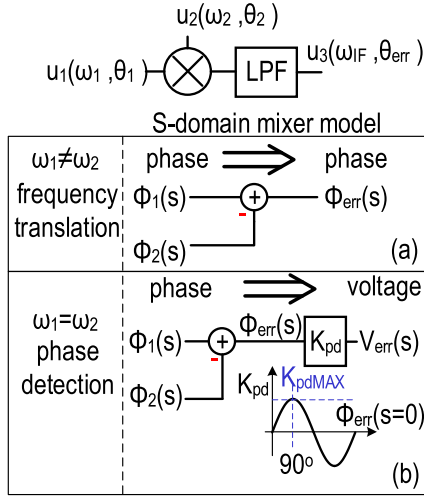


Fig. 5. Mixer for frequency translation and phase detection at (a) non-zero IF and (b) zero IF.

maximum when they are in quadrature, as shown in Fig. 5(b). In modern PDs, the mechanism is more complicated and is beyond the scope of our discussion, but the S -domain model is essentially the same.

B. Heterodyne PLL

Since the essence of phase detection is mixing, the heterodyne receiver has inspired a legacy heterodyne PLL [39] and a more recent one [40], as shown in Fig. 6. An auxiliary (AUX) clock is needed whose frequency is close to the VCO's. It mixes with the VCO signal through a mixer, frequency translating the VCO signal, and its sideband phase noise to IF. The LPF (MLPF) that follows will select the IF and suppress the undesired high-frequency tones [Fig. 5(a)]. The PD measures the phase error between transitions of CK_{FRAC} and CK_{IF} . The loop modulates the VCO phase with this phase error to force the IF feedback clock to track CK_{FRAC} . It also locks the output frequency f_{VCO2} to $f_{FRAC} + f_{AUX}$.

For the phase model, as previously discussed, the frequency translation is a subtraction function. It preserves the VCO sideband noise spectrum and transmits it with unity gain. The PD is a second subtraction block cascaded by the gain block. These two subtractions are shown in Fig. 6.

C. Fractional- N Operation in the Heterodyne PLL

There are two benefits to operate a heterodyne loop in a fractional- N mode. First, a fractional divider (N_2) outside the feedback loop lowers the DSM noise by $20 \log_{10} N_2$. In the absence of a divider in feedback, a heterodyne PLL exhibits a unity closed-loop gain and therefore does not amplify the divided DSM noise. Second, while the DSM operating frequency in a conventional loop with a feedback divider is fixed to f_{ref} which is below 100 MHz, the heterodyne loop can use a much higher DSM operating frequency f_{FRAC} to lower the in-band DSM noise and to push out the noise peaking frequency. In turn, this allows for a wide loop bandwidth. Fig. 6 shows those advantages.

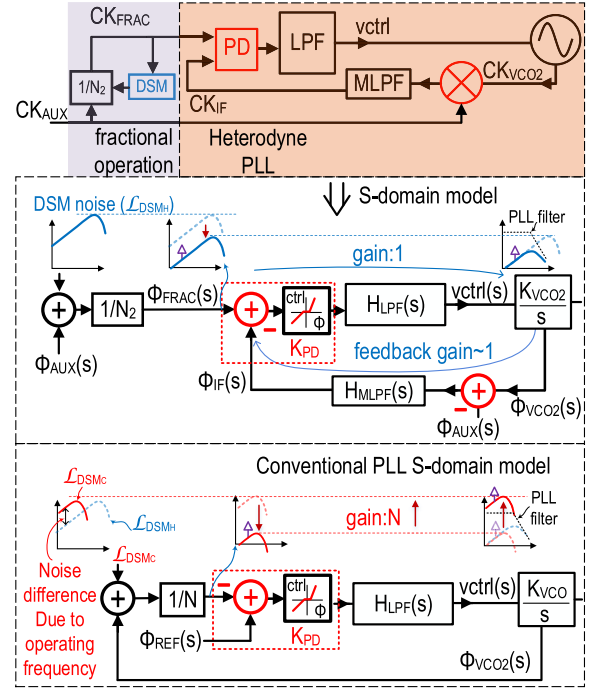


Fig. 6. Heterodyne PLL and its S -domain phase model. The DSM noise in the heterodyne PLL ($\mathcal{L}_{DSMH}$) is much lower than in the conventional loop with a feedback divider ($\mathcal{L}_{DSMC}$).

When considering PN_{in} , we normalize the DSM noise to the XTAL reference input. The DSM noise contribution in the heterodyne PLL is

$$\mathcal{L}_{DSMH, in} = \frac{\pi^2}{3 f_{FRAC} N_2^2 N^2} \left(2 \sin \left(\frac{\pi f}{f_{FRAC}} \right) \right)^{2(m-1)} \quad (9)$$

where $N = f_{VCO2}/f_{ref}$ is the overall factor of frequency multiplication and $N_2 = f_{AUX}/f_{FRAC}$. Focusing on the low-frequency noise, (9) can be simplified to

$$\mathcal{L}_{DSMH, in} \propto \frac{f^{2(m-1)}}{f_{AUX}^2 f_{FRAC}^{2m-3} N^2} \quad (10)$$

whereas the DSM noise in the conventional loop as expressed in (6) can be simplified as

$$\mathcal{L}_{DSMC, in} \propto \frac{f^{2(m-1)}}{f_{ref}^2 f_{ref}^{2m-3} N_{mmd}^2}. \quad (11)$$

Comparing (11) with (10) leads to

$$\frac{\mathcal{L}_{DSMH, in}}{\mathcal{L}_{DSMC, in}} = \left(\frac{f_{ref}}{f_{AUX}} \right)^2 \left(\frac{f_{ref}}{f_{FRAC}} \right)^{2m-3} \left(\frac{N_{mmd}}{N} \right)^2. \quad (12)$$

In mm-wave applications, f_{AUX} is in gigahertz. Thus, the first term lowers the DSM noise by more than 20 dB. A third-order DSM ($m = 3$) is often used. By pushing up f_{FRAC} to 300 MHz, the second term ensures another 28-dB noise reduction. Finally, while a conventional design uses at least an $N_{pre} = 2$ prescaler, the mixing operation in the heterodyne loop can avoid its use altogether. Then, the last term contributes another 6-dB noise reduction. In total, the heterodyne loop lowers the DSM noise by more than 54 dB, all without noise cancellation or digital calibration!

The fractional spurs introduced by the loop nonlinearity [41] are not amplified. Thus, a heterodyne PLL displays much lower spurs in its output spectrum. Like any multi-loop frequency synthesizer, the heterodyne PLL needs an extra VCO.

D. Heterodyne PLL Versus Two-Stage PLL

The third architecture in Fig. 1 also exploits the benefit of the high DSM operation frequency. The input-referred DSM noise is

$$\mathcal{L}_{\text{DSM}_T, \text{in}} \propto \frac{N_{\text{pre}}^2 f^{2(m-1)}}{f_{\text{FRAC}_T}^2 f_{\text{FRAC}_T}^{2m-3} N^2}. \quad (13)$$

Now, to assess the advantages, we compare this with the phase noise $\mathcal{L}_{\text{DSM}_H, \text{in}}$

$$\frac{\mathcal{L}_{\text{DSM}_H, \text{in}}}{\mathcal{L}_{\text{DSM}_T, \text{in}}} = \frac{1}{N_{\text{pre}}^2} \left(\frac{f_{\text{FRAC}_T}}{f_{\text{AUX}}} \right)^2 \left(\frac{f_{\text{FRAC}_T}}{f_{\text{FRAC}}} \right)^{2m-3}. \quad (14)$$

Assuming a minimum N_{pre} of 2, f_{AUX} of 4 GHz, and f_{FRAC} of $\sim 0.3\text{--}0.6$ GHz, the DSM clock rate (f_{FRAC_T}) in the conventional two-stage architecture needs to be $3\times$ faster than the heterodynes to reach the same DSM noise. This will inevitably consume more power and require a more linear PD. The conventional second stage still has at least 20-dB closed-loop gain and thus introduces much higher loop component noise than the heterodyne loop. The heterodyne architecture is, therefore, a much better choice than the conventional two-stage design.

E. Under-Sampling in PLL

An AUX clock is required to achieve the heterodyne PLL operation, which, in principle, can be generated through a separate on-chip integer- N PLL (AUX PLL). However, choosing f_{AUX} to be close to the mm-wave frequency f_{VCO_2} has three practical drawbacks.

- 1) An AUX VCO at mm-wave frequency will suffer from lower FoM_{VCO} .
- 2) Distributing the AUX clock to the heterodyne loop (or main loop) can be power-hungry.
- 3) The two oscillators may experience unwanted mutual pulling [42], [43].

An HMPLL can avoid these problems [22], [23], [24], [25]. The mixer in the feedback loop is replaced by a sampler, as shown in Fig. 7. The edge sampling generates an impulse train in the frequency domain that convolves with the VCO spectrum. If the N_{AUX} th harmonic of the impulse train is close to the VCO frequency, it will translate the VCO spectrum to IF where $f_{\text{IF}} = f_{\text{VCO}_2} - N_{\text{AUX}} f_{\text{AUX}}$. This is the aliasing that results from under-sampling (sub-sampling) [44].

At offset frequencies higher than 100 MHz, the output phase noise of PLL is often negligible. It can be considered a band-limited noise whose double-sided bandwidth, defined as f_{PN_BW} in Fig. 7, is twice this maximum offset frequency. The VCO spectrum is a tone at f_{VCO_2} with a band-limited phase noise surrounding it. The under-sampling shifts the entire VCO spectrum to IF. The Nyquist sampling theorem teaches that by keeping the sampling frequency twice as high as the phase

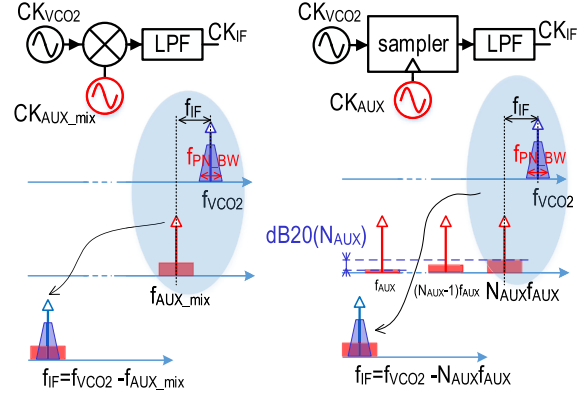


Fig. 7. Harmonic mixing versus direct mixing.

noise bandwidth, the waveform of phase noise is preserved. Between dc and f_{VCO_2} , the additive thermal noise is much lower than the phase noise and can be neglected. Thus, the minimal f_{AUX} needs to be higher than $2f_{\text{PN}_\text{BW}}$, which is roughly 400 MHz. Since the aliased IF tone is located at the center of the phase noise, f_{IF} should at least be higher than $f_{\text{PN}_\text{BW}}/2$, which is roughly 100 MHz. These constraints are easily satisfied. In fact, f_{AUX} is chosen mainly to lower the spectral density of DSM noise, which will be discussed in Section IV. In our circuit, f_{AUX} is roughly 4 GHz.

By contrast, the sub-sampling PLL [45] uses the corresponding harmonic of f_{ref} to frequency translate the VCO spectrum to dc. It merges the IF frequency translation and phase detection in the HMPLL into one step and ensures a unity closed-loop gain. Unfortunately, the need to equate Nf_{ref} with f_{VCO} imposes a limitation in fractional operation. Without the correct edge alignment, the subsampling PD gain varies significantly [Fig. 6(b)], which disturbs loop operation. A DTC can modulate the reference to align the sampling edge with the zero crossing of the VCO [46]. However, this introduces extra phase noise, degrading PN_{in} , and requires digital calibration of gain mismatch, slowing the settling time. In the HMPLL, translation to the IF frequency does not need edge alignment. Instead of measuring phase difference at zero IF by the error from zero crossing in samples of an oscillating waveform's voltage—where the gain of phase detection depends on waveform shape—at nonzero IF phase detection reverts to the conventional method of measuring phase difference with a pulsewidth-modulated square wave defined by consecutive rising or falling edges of the reference and translated VCO oscillations. This PD inherently shows a wide linear operating range, thus maintaining constant loop gain over the jumps that appear in edge timing as the fractional divider switches modulus to generate CK_{FRAC} .

Similar to the VCO phase noise, which scales with operating frequency, phase noise in the AUX path rises with the harmonic index [45]. Thus, when using N_{AUX} th for harmonic mixing, the AUX clock phase noise has increased by $20 \log_{10} N_{\text{AUX}}$.

F. HMPLL Model

Compared with the heterodyne PLL model, the only difference in HMPLL is the gain of N_{AUX} from AUX PLL to the

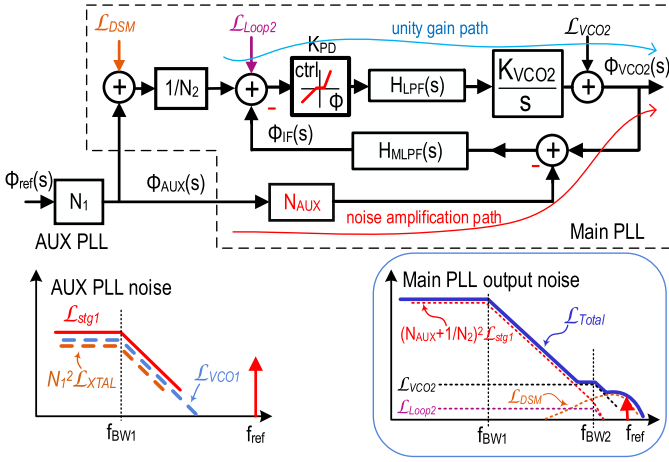


Fig. 8. HMPLL phase model.

main loop in Fig. 8. As discussed, phase noise $\mathcal{L}_{loop2}$ arising from the main PLL's components experiences no amplification. The DSM noise $\mathcal{L}_{DSM}$ is lowered dramatically. The main loop, now designed with wide bandwidth, suppresses the phase noise of the mm-wave VCO $\mathcal{L}_{VCO2}$. The AUX PLL is an integer- N architecture that multiplies the xtal reference into the gigahertz range. However, it also amplifies the xtal phase noise $\mathcal{L}_{XTAL}$ and introduces its own phase noise dominated by the AUX VCO $\mathcal{L}_{VCO1}$. The AUX loop noise is transmitted through the main loop via two paths: the harmonic-mixing path N_{AUX} and the fractional divider path $1/N_2$ (Fig. 8). The noise from the two paths adds correlatively. Thus, the AUX loop noise is amplified by $(N_{AUX} + 1/N_2)$ and determines the final output phase noise. The new IPN is approximately

$$\langle \phi_{total, opt}^2 \rangle \approx \frac{2\pi f_{VCO2}^2}{f_{ref}} \sqrt{\frac{10^{-\frac{FoM_{VCO_{AUX}}}{10}} PN_{in_{AUX}}}{P_{VCO_{AUX}}/1 \text{ mW}}}. \quad (15)$$

Although f_{VCO2} is in the mm-wave region, $FoM_{VCO_{AUX}}$ and $PN_{in_{AUX}}$ originate in a sub-10-GHz integer- N PLL. Both are much superior to a single-stage mm-wave synthesizer. The XTAL reference noise is amplified by $N_1(N_{AUX} + 1/N_2)$, which is exactly the overall frequency multiplication ratio f_{VCO2}/f_{ref} . This scaling is a fundamental property of frequency multiplication. Finally, the two cascaded loops strongly filter any spurs at multiples of f_{ref} .

IV. COMPLETE ARCHITECTURE AND FREQUENCY PLAN

In Section III-F, we have shown that the AUX integer- N PLL noise dominates the final phase noise at the synthesizer's output. Based on (15), the AUX VCO FoM_{VCO} is critical to overall performance. Simulations indicate that the VCO LC tank's true quality factor Q_{tru} [47] peaks at around 10 GHz. Thus, as shown in the frequency plan of Fig. 9, the AUX PLL outputs a sub-10-GHz clock for best FoM_{VCO} and for acceptable power consumed in the VCO buffer and divider. The cascaded frequency divide-by-two keeps in check power lost driving the parasitics of the long metal route from CK_{AUX} to the following sampler and in the divider N_2 . Further lowering of f_{AUX} will not yield significant savings in power.

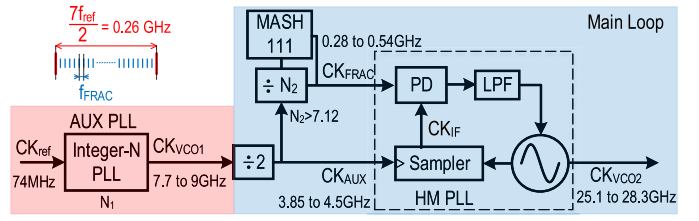


Fig. 9. Overall synthesizer frequency plan.

Meanwhile, a ~ 4 -GHz clock will efficiently lower the DSM noise as indicated in (10). Also, Section VII-A shows that such f_{AUX} can effectively avoid mixing spurs. With all those considerations, the AUX clock frequency and the range of N_{AUX} are determined. The 7th harmonic of f_{AUX} frequency translates the mm-wave VCO output to IF. This sets the coarse frequency tuning step of the whole synthesizer to $7f_{ref}/2$, which is 260 MHz in our case. CK_{FRAC} must interpolate this coarse step, and thus, its tuning range is set to 0.28–0.54 GHz. The near-0.3-GHz minimal operating frequency also significantly reduces the DSM noise, as (10) has described. From Fig. 9, the output frequency can be expressed as

$$f_{VCO2} = N_1 \frac{7f_{ref}}{2} + N_1 \frac{f_{ref}}{2} \frac{1}{N_2}. \quad (16)$$

As for the fractional divider N_2 , the DSM modulates its phase similar to the conventional fractional- N PLL. Thus, there is no undesired consequence of the nonlinear relationship between f_{VCO2} and N_2 .

In summary, the main properties of this new HMPLL architecture are given as follows.

- 1) The unity closed-loop gain of the main PLL does not raise DSM noise contribution and DSM-related fractional spurs, allowing a wide bandwidth loop design to suppress the mm-wave VCO noise effectively. This stage resembles an ideal fractional frequency multiplier.
- 2) The complete PLL noise tradeoff is as efficient and straightforward as in a simple sub-10-GHz integer- N PLL, which enjoys the best PN_{in} and FoM_{VCO} , overcoming the performance limitation due to the limited FoM_{VCO} in a single-stage mm-wave PLL. This accords with the discussion in Section II.

V. COUPLED MM-WAVE VCO

Although the wideband HM-PLL strongly filters the mm-wave VCO phase noise, its bandwidth is not infinite, and improving mm-wave VCO phase noise is always beneficial.

To improve phase noise, one can increase the LC tank power or improve its Q_{tru} [48], [49]. When the supply is fixed, lowering the inductance is the only way to increase the tank power.¹ As noted in [49], there is a limit on the minimal size of a spiral inductor before Q_{tru} drops.

¹More precisely, phase noise is lowered proportionally with the effective series resistance that captures inductor loss at the oscillation frequency. The smaller the trace length of sufficient width, the lower this resistance. To lower it below the length of one turn, two or more single-turn spirals can be connected in parallel.

	(a) 2-port	(b) 2-port shrink	(c) lightly coupled 4-port	(d) twisted 4-port	(e) 2-port bowtie
Equivalent Inductance	100pH	50pH	46pH	50pH	52pH
Q_{tru}	29	19	27	29	10
Tank Current	Low	High	High	High	High
Oscillation Mode	One	One	Two	One	One
Magnetic Isolation	Low	Low	Low	High	High
Spiral Diameter	86um	50um	77um	72um	33um

Fig. 10. Different inductor structures and their property.

	Twisted best location	Twisted sub-optimal location	Twisted worst location	Non-Twisted	lightly coupled best location
distance (um)					
Isolation (dB)					
240	-56	-45	-26	-12	-18
320	-60	-49	-28	-18	-24
400	-63	-50	-30	-24	-29
480	-65	-50	-32	-28	-33

Fig. 11. Magnetic isolation comparison for different structures and locations.

Although two lightly coupled inductors [50] as shown in Fig. 10(c) can lower the total inductance by almost $2\times$ without degrading Q_{tru} , they enable multiple resonant modes. Following the concept of [49], the two inductors can be merged into one twisted four-port inductor [Fig. 10(d)]. Now, the current flow is forced to follow the “8” shape and selects one mode unambiguously. The ac current behavior is the same as a bowtie inductor [Fig. 10(e)], which is known to display good magnetic immunity [51], [52], [53].

EMX simulation (Fig. 10) reveals that the twisted four-port displays more than $2\times$ higher Q_{tru} than the corresponding two-port bowtie inductor. Another EMX simulation (Fig. 11) to explore coupling to nearby spirals shows that the twisted four-port can provide >45 -dB increase in isolation and is not sensitive to position: a 30° misalignment degrades the isolation by only 10 dB. Although one of the modes of the lightly coupled four-port demonstrates good magnetic isolation, given the difficulty in enforcing the desired mode of oscillation we prefer to use the twisted four-port geometry.

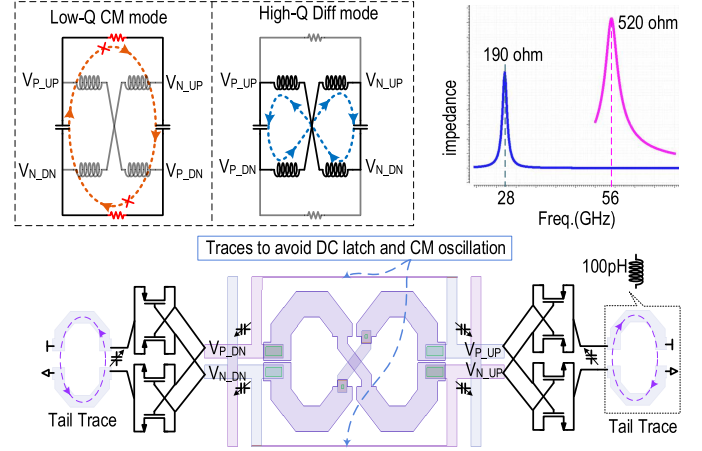


Fig. 12. Coupled mm-wave VCO.

In the actual design (Fig. 12), two thin traces bypass the inductor to generate a low- Q dc path that suppresses latching the differential pairs. At the wanted frequency, the added high-resistive CM mode line cannot sustain oscillation, ensuring that the VCO only oscillates in the high- Q differential mode.

In each active core, two lightly coupled metal traces form a 100-pH inductor in the tail of the differential pair, resonating with the small parallel capacitor bank at $2f_{VCO2}$. The resonant impedance is around $500\ \Omega$, which is even higher than that of the main tank. As shown in [54] and [55], this significantly avoids degradation of main tank loaded Q_{tru} , thus lowering the oscillator's phase noise. In the mm-wave range, a well-designed unit capacitor with an ON-OFF ratio of 2 exhibits Q of only 30. To resonate with the 55-pH inductor across the frequency tuning range of 25–28 GHz, the capacitor bank should be made to vary by 150 fF. The switch losses associated with the large bank of unit capacitors degrade the main tank Q_{tru} significantly. In any case, Q 's of both varactor and lossy FET capacitance are lower in this frequency range, further degrading the main tank Q_{tru} . To cover this tuning range, the simulated FoM_{VCO} never exceeds 190 dB.

VI. AUX PLL DESIGN

As mentioned in Section IV, the integer- N PLL, which generates the AUX clock ultimately, limits the overall performance. Its PN_{in} needs to be as low as possible.

The AUX clock can be synthesized in a subsampling PLL. This has no divider noise, and its large PD gain suppresses the LPF noise. However, it has a very narrow locking range that requires an extra conventional loop to assist in frequency locking. Also, the VCO is prone to reference kickback spurs.

On the other hand, an AUX clock synthesized in a sampling loop has a much wider lock range [56]. This loop is shown in Fig. 13(top). This too offers a large PD gain to suppress the LPF noise [57]. A retimer after the divider resets the noise and confines the divider noise ($\mathcal{L}_{N1}$) contribution to just one flip-flop, which is much lower than $\mathcal{L}_{XTAL}$.

We use a sampling PLL. It has a similar noise performance to a subsampling PLL but is more robust.

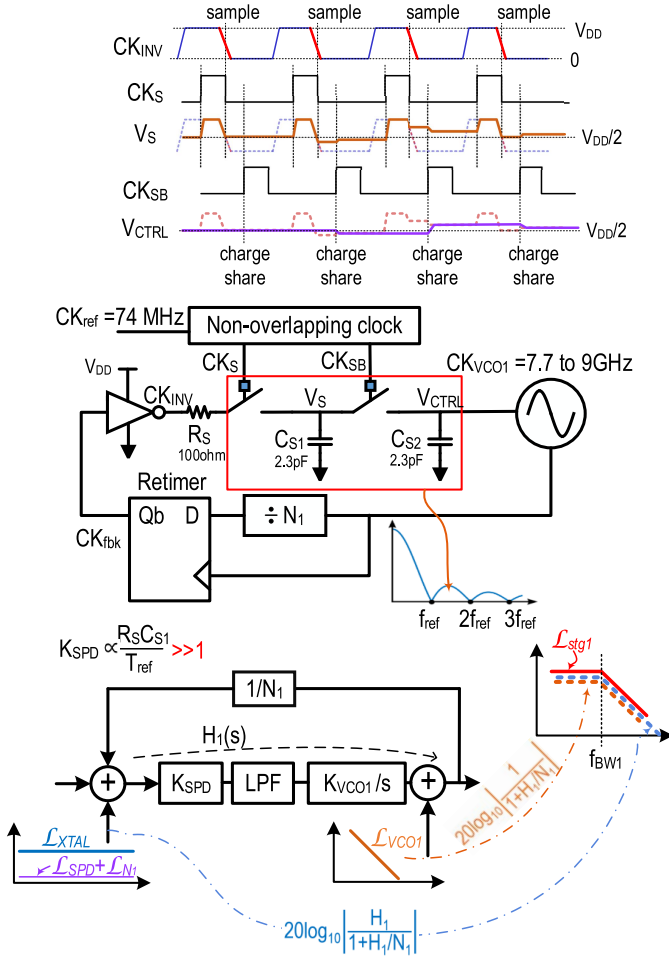
Fig. 13. Integer- N PLL structure and model.

Fig. 13 shows the operation of the sampling PD. CK_S samples the inverter's falling edge, which contains the phase information. The sampled charge is shared with C_{S2} and is held there for one reference cycle. R_S is added such that the time constant $R_S C_{S1}$ mainly determines the falling time of the inverter and the gain of the sampling PD (K_{SPD}) [13], desensitizing the gain to transistor process variation

$$K_{SPD} = \frac{T_{ref}}{2\pi} \frac{V_{DD}}{t_{fall}} \propto \frac{T_{ref}}{R_S C_{S1}}. \quad (17)$$

R_S of only 100 Ω dominates the inverter self ON resistance yet introduces negligible noise and maintains a large K_{SPD} . When input-referred, the equivalent phase noise from the PD ($\mathcal{L}_{SPD}$) is much lower than $\mathcal{L}_{XTAL}$. The PD works at a sub-100-MHz frequency. Even when loaded by a large capacitor C_{S1} , it consumes less than 100 μA .

The mentioned charge sharing between C_{S1} and C_{S2} forms a switch-capacitor LPF with an equivalent transfer function of [56]

$$H_{LPF_S}(s) = \frac{1}{1 + \frac{C_{S2}}{C_{S1}} \frac{s}{f_{ref}}}. \quad (18)$$

C_{S2} holds the charge for one period cycle, which realizes a transfer function $\text{sinc}(f/f_{ref})$ that nulls transmission at f_{ref}

and harmonics. Due to the large K_{SPD} , the noise from the switch-capacitor filter becomes negligible when input-referred. The preferred V_{CTRL} range is between 0.3 and 0.5 V, where K_{SPD} experiences $\pm 30\%$ variation.

This architecture is power-efficient and offers low PN_{in} and low reference spurs. The AUX VCO uses a tail inductor [54], [55] to avoid degrading Q_{tru} of the main tank. Due to the choice of VCO frequency, the oscillator achieves a good FoM_{VCO} of around 191 dB, leading to outstanding overall PLL performance.

VII. HMPLL DESIGN

A. Sampler Design

The sampler in the HMPLL, as shown in Fig. 14, is very similar to that used by the AUX loop sampling PD. Since the VCO buffer is overdriven, it introduces odd harmonics at its output. They will mix with harmonics contained in the train of sampling impulses. The mixing products can be any combination of $Mf_{VCO2} - M_{AUX}f_{AUX}$ where M is an odd integer, but M_{AUX} , which is a harmonic of the sampling impulse, can be any integer value. One concern is that some combinations can introduce a mixing spur close to the IF and become an unwanted in-band spur. Since the loop bandwidth is within 10 MHz, this happens when

$$f_{spur} = |Mf_{VCO2} - M_{AUX}f_{AUX} - f_{IF}| \leq 10 \text{ MHz}. \quad (19)$$

When $Mf_{VCO2} - M_{AUX}f_{AUX} \geq 0$ and recalling from Section IV the choice of the 7th harmonic, this inequality can be expressed as

$$(7M - M_{AUX}) + \frac{M-1}{N_2} \leq \frac{10 \text{ MHz}}{f_{AUX}}. \quad (20)$$

Otherwise, (19) is expressed as

$$(M_{AUX} - 7M) - \frac{M-1}{N_2} \leq \frac{10 \text{ MHz}}{f_{AUX}}. \quad (21)$$

Based on our frequency plan in Section IV, f_{AUX} is around 4 GHz and $10 \text{ MHz}/f_{AUX}$ is less than 3/1000. The fractional division N_2 is between 7.12 and 16.

If $7M - M_{AUX} = 0$, (19) holds only when $M = 1$ and $Mf_{VCO2} - M_{AUX}f_{AUX}$ creates only the wanted tone at IF but no spurs.

If $7M - M_{AUX} = \pm 1$, $M = 11$ is the smallest odd harmonic of f_{VCO2} that will satisfy (19). The corresponding M_{AUX} is 78 and N_2 is set between 9.97 and 10.03. For any smaller M , the corresponding N_2 is outside our design range.

If $7M - M_{AUX} \geq \pm 2$, $|M|$ must be larger than 14 so that N_2 is within the designed range. However, the amplitude of higher harmonics is always weaker. Therefore, $M = 11$ is the worst case.

$11f_{VCO2}$ and $78f_{AUX}$ are located close to 300–350 GHz. As shown in Fig. 14, the Thevenin equivalent of the inverter and the first switch of the sampler is a square-wave voltage in series with an ideal switch and a resistor R_{ON} which consists of both inverter ON resistance and switch ON resistance. R_{ON} and $C_{sampler1}$ form an LPF, suppressing the harmonics at $11f_{VCO2}$ and higher [58]. It passes f_{VCO2} to capture the wanted VCO frequency. Thus, this architecture requires no prescaler.

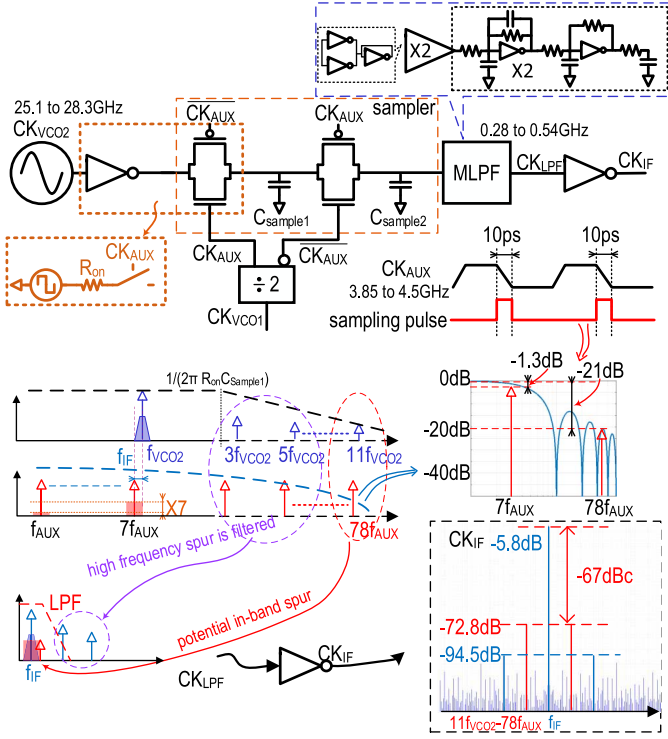


Fig. 14. HMPLL sampler's schematic, frequency-domain operation, and the simulation for the worst case mixing spur.

A practical sampling pulse in this technology will have fall time higher than 10 ps. As shown in Fig. 14, such a sampling pulse will lower the magnitude of the high-frequency sampling harmonics by at least 21 dB while only lowering the sampling strength of the 30 GHz tone by -1.3 dB. Thus, the sampling itself helps reduce the high-frequency mixing spurs by at least ~ 20 dB. Our simulations confirm that the final in-band mixing spur level is -67 dBc. In measurement, this mixing spur cannot be observed because the frequency of the related tones is so high that it will be strongly attenuated by parasitic capacitance.

$Mf_{VCO2} - M_{AUX}f_{AUX}$ also generates high-frequency tones. The filter (MLPF), following the sampler and sharing a similar design as [24], effectively suppresses them.

The sampling pulse leads to a conversion gain G_{mix} of -1.3 dB. The intrinsic LPF formed by $R_{ON}C_{sampler1}$ attenuates the signal by G_{atn} , which is around -5 dB based on our simulation. Similar to the switch-capacitor filter in our sampling PLL, the two switched capacitors also form an LPF

$$H_{SC}(s) = \frac{1}{1 + \frac{C_{sampler2}}{C_{sampler1}} \frac{s}{f_{AUX}}} \quad (22)$$

$C_{sampler1}$ is around 20 fF to reduce buffer power, while $C_{sampler2}$ is around 100 fF due to the MLPF parasitic. Given that f_{AUX} ranges across 3.85 to 4.5 GHz, the pole of H_{SC} is between 120 and 140 MHz, which introduces another 7–14-dB attenuation. The initial strength of the signal is around -6 dBV. After all the attenuations, the final signal level at the MLPF input is $-6 + 20 \log_{10}(|G_{atn}G_{mix}H_{SC}(f_{IF})|)$, which is roughly -19 to -26 dBV. At this input level, the

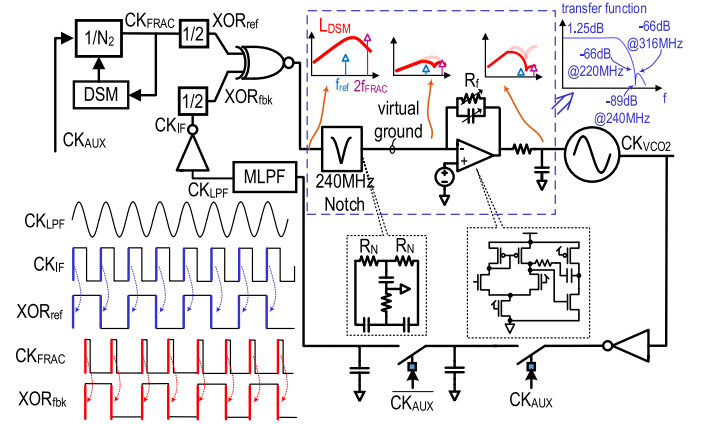


Fig. 15. Architecture of main PLL.

MLPF functions as a linear filter and does not introduce noticeable intermodulation spurs. The input-referred noise of MLPF is dominated by its first stage, which consumes 0.2 mA and has a transconductance (g_{m1}) of 4 mS. Thus, $4KT\gamma/g_{m1}$ is around -177 dBV/Hz. The phase noise introduced by the MLPF is less than -150 dBc/Hz. Due to the unity closed-loop gain, phase noise at the final PLL output remains about -150 dBc/Hz. This is very low.

B. Main PLL Loop Design

Fig. 15 shows the complete block diagram of the main loop. A divide-by-two is in series with the XOR-based PD. It samples the rising edge of CK_{FRAC} and provides the PD a 50% duty-cycle input. Since the DSM modulation is all in the rising edge of CK_{FRAC} and XOR PD utilizes both rising and falling edges of its input for phase detection, this architecture captures all the edge information and ensures no spectrum folding of the DSM noise. The XOR-based PD is linear and has a wide operating range to tolerate the modulated clock edge of CK_{FRAC} . Referred to its input, the added divide-by-two amplifies the noise of PD and the following LPF by 6 dB only. The DSM noise and the harmonic-mixing chain still experience 0-dB gain. The dedicated fractional- N loop of [22] is eliminated, saving area and power. The large voltage swing from the peak of the DSM noise and the reference spurs of the XOR output may overdrive the active LPF. A passive low- Q twin-T notch suppresses the DSM peaking noise and the high-frequency reference spurs of the second loop. The RC filter on the opamp feedback path provides extra high-frequency filtering. Overall, the amplifier—a conventional two-stage single-ended opamp—experiences minimal swing at both its input and output and is thus very linear. The programmable feedback resistor and the notch resistance determine the tunable dc gain of the filter. With a supply of V_{DD} , the loop unity-gain frequency is simply

$$f_{u_{HMPLL}} = \frac{V_{DD}K_{VCO2}R_f}{\pi 4R_N} \quad (23)$$

The notch is set to 240 MHz and introduces negligible phase shift below 10-MHz offset. Therefore, the loop can push

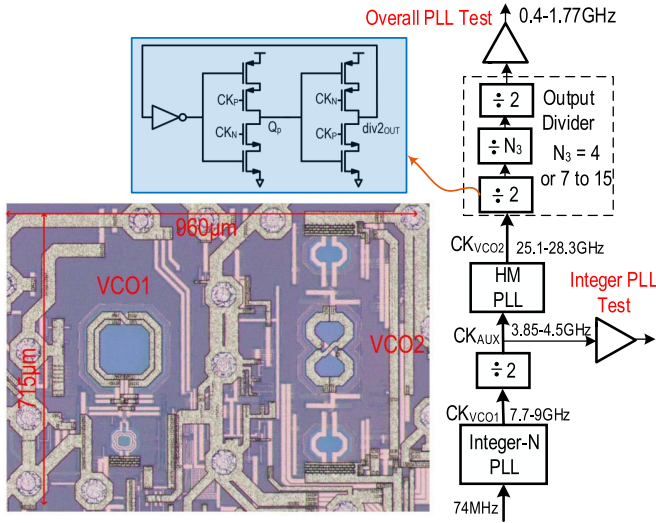


Fig. 16. Die photograph and test strategy for the whole PLL.

the bandwidth to as high as 5–10 MHz without introducing peaking in the transfer function.

VIII. MEASUREMENT

The chip is fabricated in TSMC 7-nm CMOS. Fig. 16 shows our test method and the chip die photograph. A test buffer is placed to measure the AUX clock's phase noise directly. The synthesizer output frequency is around 30 GHz. Instead of direct measurement, a tunable output divider N_3 is designed to lower the frequency to below 2 GHz, relaxing the PCB design, which transmits the waveform to a spectrum analyzer. The phase noise and spur are measured at the <2 GHz output and then scaled to the corresponding PLL output frequency based on the dividing ratio of N_3 .

Fig. 17 shows the measurement results of the AUX clock. This PLL, in total, consumes 6 mW. The AUX VCO and its buffer burn 5.2 mW. The divider N_1 and the sampling PD, in total, burn 0.8 mW. The AUX PLL creates 76-fs rms jitter. Due to the intrinsic sinc function, the reference spur is only -74 dBc when scaled to 8-GHz output. This is much lower than that in [56] because our optimal loop bandwidth (~ 1 MHz) is much smaller than that of [56] (~ 9 MHz) and our reference spur offset frequency (~ 74 MHz) is more than three times higher than that in [56] (~ 22.6 MHz). Table I shows that both FoM_{PLL} and $\text{FoM}_{\text{PLL},r}$ are among the best in the-state-of-art, confirming that a sub-10-GHz synthesizer approaches the optimum performance in this technology.

Next, we present the performance of the mm-wave VCO. Table II shows its measured phase noise. Consuming 3.2 mW, it has a tuning range of 25.1–28.3 GHz and achieves 188-dB FoM_{VCO} , which is also among the best.

Fig. 18 shows the phase noise of the overall synthesizer, which matches the prediction almost exactly. The integrated rms jitter, including all the coupling spurs, is 88 fs. This verifies that the second loop contributes very little noise. In total, the main PLL consumes only 5.8 mW. Inside it, the harmonic-mixing block and MLPF consume 0.96 mW, and the

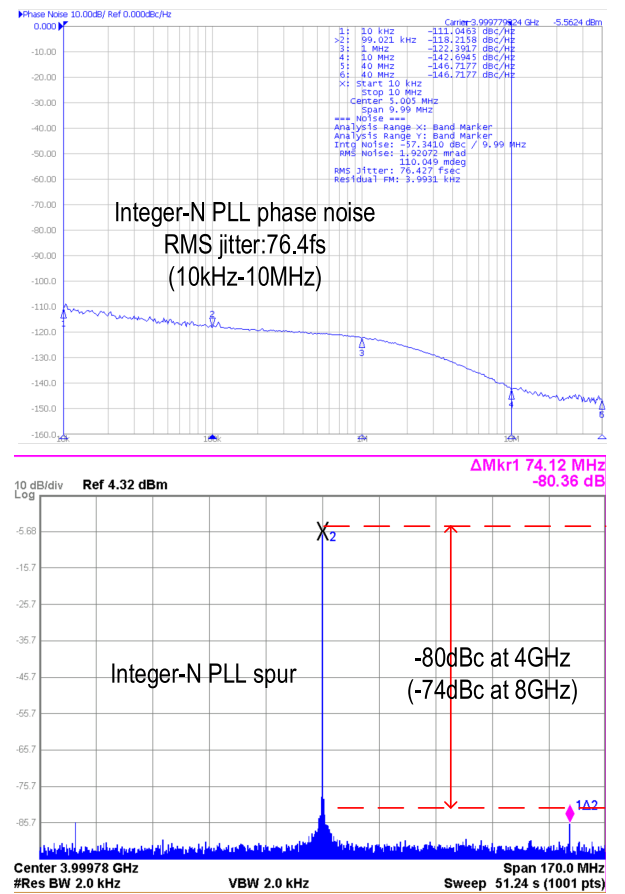


Fig. 17. AUX PLL measurement at 4 GHz.

TABLE I
INTEGER- N PLL TABLE OF COMPARISON

	This work	Y.Lim ISSCC20	Z. Yang ISSCC19	Z. Zhang ISSCC19	X. Gao ISSCC09	A. Sharkia ISSCC18
Process	7nm CMOS	65nm CMOS	65nm CMOS	40nm CMOS	0.18μm CMOS	65nm CMOS
$f_{\text{out}}[\text{GHz}]$	7.5-9	12-14.5	25.4-29.5	12-16	2.2	5
$f_{\text{ref}}[\text{MHz}]$	74	50	103	200	55	100
RMS jitter[fs]	76.6	83	71	56	150	185.3
Ref Spur[dBc] (normalized to 8GHz)	-74	-79.54	-73.3	-70	-35	-60.1
Power[mW]	6	6.7	10.2	7.2	7.6	1.1
FoM	-254.5	-253	-252.9	-256	-247.8	-254
FoM_r	-245.8	-246	-242.7	-243	-240.4	-244
Aare(mm ²)	0.126	0.23	0.24	0.23	0.18	0.01
$\text{FoM} = 10\log_{10}(\text{jitter}^2) + 10\log_{10}(\text{P}_{\text{DC}}/1\text{mW})$						
$\text{FoM}_r = 10\log_{10}(\text{jitter}^2) + 10\log_{10}(\text{P}_{\text{DC}}/1\text{mW}) + 10\log_{10}(f_{\text{ref}}/10\text{MHz})$						

loop LPF filter consumes only 0.2 mW. The CK_{AUX} buffer, which drives a long line, dissipates 1.1 mW. The two-stage PLL consumes 12.9 mW and achieves -250-dB FoM_{PLL} . Due to the sinc filtering of AUX PLL and the cascade filter of the main loop, the 74-MHz reference spur cannot be detected at the final output, while the reference spur of the main loop is only -83 dBc when normalized to 28-GHz output.

TABLE II
MM-WAVE VCO TABLE OF COMPARISON

	This work	D. Murphy JSSC18	Z. Zong JSSC16	H. Guo ISSCC19	H. Jia ISSCC21
Current[mA]	4	17	13.5	3.8-4	40.9-42.9
Supply[V]	0.8	0.95	0.7	0.48	0.55
PN @1M (normalized to 28.3G)	-103.4 to -104.58	-108.9	-105.5	-108.33	-111
Frequency [GHz]	25-28	23-29.9	16.1-20.8	25.48-29.92	52.4-60.4
Tuning range(TR) [%]	11.3	26	25.4	16	14.2
FoM (thermal region)	188.71-189.2	186.5-187	184.5-186.2	189.8-191.6	182.2-186.5
Flicker corner (kHz)	300-500	300-500	950	130-230	NA
FoM _T (thermal region)	189.77-190.2	194.6-195.3	192.6-193.4	193.9-195.7	185.2-189.5
Techonology	7nm CMOS	40nm CMOS	40nm CMOS	65nm CMOS	65nm CMOS
Area(mm ²)	0.065	0.1	0.07	0.08	0.032
FoM=[PN]+20log ₁₀ (f ₀ /Δf)-10log ₁₀ (P _{DC} /1mW)		FoM _T =FoM+20log ₁₀ (TR/10)			

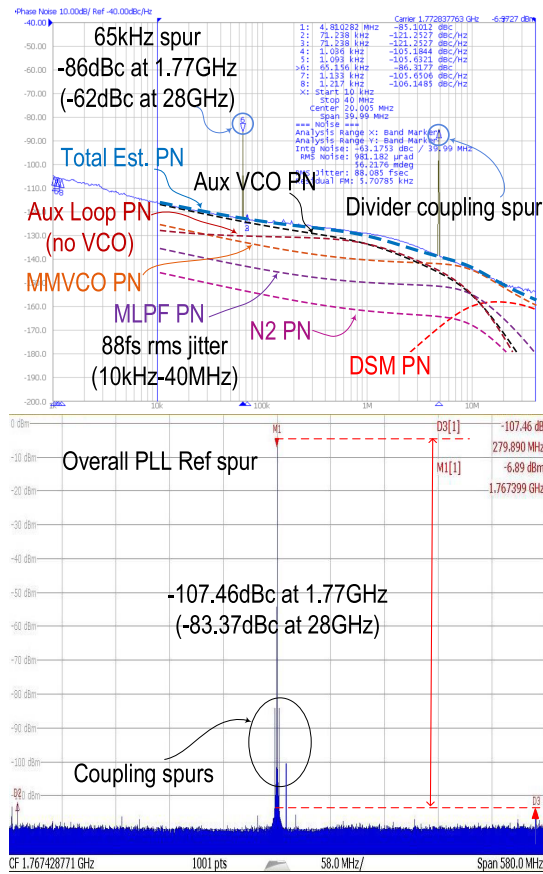


Fig. 18. Overall phase noise: measurement versus prediction, Ref spur level.

The overall noise plot in Fig. 18 (top) shows two spurs. The 65-kHz spur is not related to the frequency or the divide ratio but is probably coupled with the supply or JTAG clock. The 4.8-MHz spur is caused by coupling between the output test buffer and the reference buffer, both connected to the same supply on the PCB without an internal LDO. A change in ratio N_3 of the output divider, as shown in Fig. 19(a) and (b), shifts this spur. This confirms that it is not a PLL fractional spur: adding a dedicated LDO should eliminate it. The worst

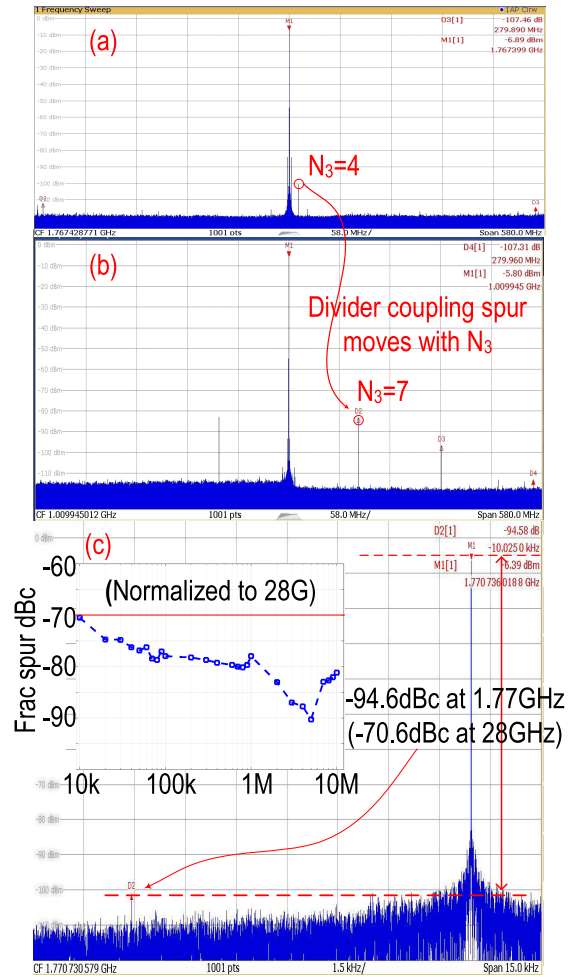


Fig. 19. (a) and (b) Output divider coupled spur change with dividing ratio. (c) Overall PLL fractional spur measurement.

TABLE III
MM-WAVE PLL TABLE OF COMPARISON

	This Work	L.Grimaldi ISSCC19	H.Yoon ISSCC18	M. Ferriss ISSCC15	D. Weyer ISSCC18	J.Kim ISSCC19
Mode	Frac-N	Frac-N	Frac-N	Frac-N	Frac-N	Integer-N
Process	7nm CMOS	65nm CMOS	65nm CMOS	32nm SOI	40nm CMOS	65nm CMOS
f _{out} [GHz]	25 to 28	30.6 to 34.2	25 to 30	13.1 to 28	36.3 to 38.2	28 to 31
f _{ref} [MHz]	74	100	120	104	120	100
rms jitter[fs]	88	197.6	206	1027	1150	76
Frac Spur[dBc] (normalized to 28G)	<-70 ³	-43	n.a.	n.a.	-57.6	n.a.
Ref Spur[dBc] (normalized to 28G)	<-83	n.a.	-83	-66	-57.6	-58
Power[mW]	12.9	35	36.4	31	68	41.8
FoM ¹	-250	-238.6	-235.1	-225	-220.5	-246.1
FoM _T ²	-241	-228.6	-224.1	-214	-209.5	-236.1
Core Area(mm ²)	0.24	0.55	0.95	0.24	0.18	0.32
1: FoM = 10log ₁₀ (jitter ²)+10log ₁₀ (PDC/1mW)						
2: FoM _T = 10log ₁₀ (jitter ²)+10log ₁₀ (PDC/1mW)+10log ₁₀ (f _{ref} /10MHz)						
3: Ignored coupling spurs						

fractional spur is also swept and remains below -70 dBc, as shown in Fig. 19(c). This confirms that the second loop is linear enough to avoid any significant fractional spurs. We can conclude that the fractional- N spur level is lower than -70 dBc.

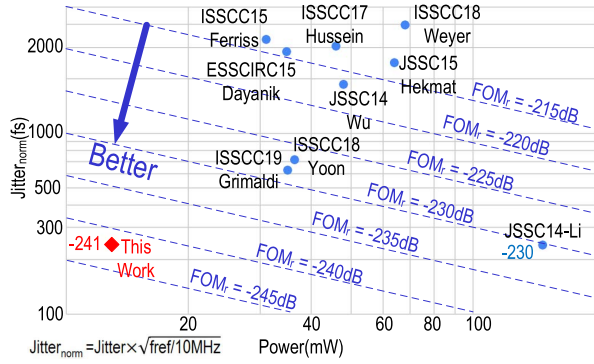
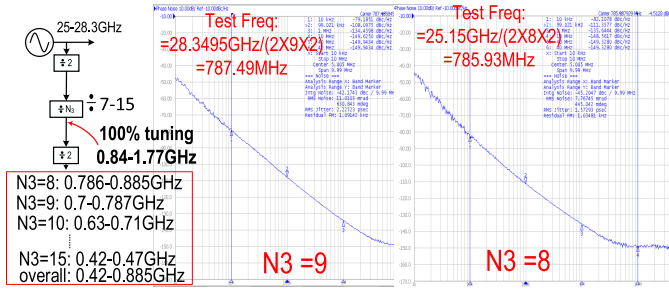
Fig. 20. mm-wave fractional- N PLL FOM_r comparison.

Fig. 21. Direct division of a high-frequency VCO to achieve 100% tuning.

Table III compares our PLL performance with the state of the art. The presented PLL's FoM and FoM_r are 10 dB better than those of the recent fractional- N mm-wave PLL. It is even 5 dB better than a recently published integer- N PLL [59]. The mm-wave fractional- N PLL FoM_r survey in Fig. 20 illustrates the leap in performance of our architecture more clearly. While most of the recent publications are in the -220 -dB region, our PLL is the first to reach -241 dB with a sub-100-MHz reference. This is comparable with the state-of-the-art sub-10-GHz fractional- N synthesizer shown in Fig. 4. In this prototype, the coarse frequency calibration is not implemented. Any open-loop frequency calibration [60], [61], [62] is suitable for this PLL. When calibration is done, the coarse locking circuit is turned off to avoid extra power consumption. The PLL can adjust the residual frequency error.

A high-frequency VCO allows us to explore 100% tuning range operation for software-defined radio (SDR). SDR is a versatile radio platform that can be suitable for different wireless standards [63]. Such a platform needs to cover a vast frequency range to support different modes. Thus, a 100% tuning range oscillator is a core for such an implementation. Multiresonant modes of an oscillator [64], [65] or multiple LC tanks with different resonant frequencies have been explored to achieve this function. Those methods either degrade the VCO FoM_{VCO} or occupy a large area. With a 30-GHz clock, as shown in Fig. 21, by a simple frequency division, a 0.8–1.8-GHz 100% frequency tuning clock is achieved without large chip area or VCO noise degradation over a large tuning range.

IX. CONCLUSION

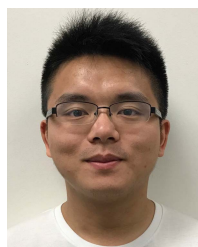
This article presents a two-stage HMPLL architecture with a coupled mm-wave VCO. The advantage of this new architecture is supported by the effective filtering of the mm-wave VCO noise and DSM quantization noise. Furthermore, it is power-efficient and immune to spurs. This is the first reported mm-wave PLL architecture that can reach -241 -dB FoM_{PLLr} .

REFERENCES

- [1] Z. Deng and A. M. Niknejad, "A 4-port-inductor-based VCO coupling method for phase noise reduction," in *Proc. IEEE Custom Integr. Circuits Conf.*, Sep. 2010, pp. 1–4.
- [2] Z. Zong, M. Babaie, and R. B. Staszewski, "A 60 GHz frequency generator based on a 20 GHz oscillator and an implicit multiplier," *IEEE J. Solid-State Circuits*, vol. 51, no. 5, pp. 1261–1273, May 2016.
- [3] B. Sadhu et al., "A linearized, low-phase-noise VCO-based 25-GHz PLL with autonomic biasing," *IEEE J. Solid-State Circuits*, vol. 48, no. 5, pp. 1138–1150, May 2013.
- [4] Y. Hu, T. Siriburanon, and R. B. Staszewski, "A 30-GHz class-F23 oscillator in 28 nm CMOS using harmonic extraction and achieving 120 kHz $1/f^3$ corner," in *Proc. 43rd IEEE Eur. Solid State Circuits Conf. (ESSCIRC)*, Sep. 2017, pp. 87–90.
- [5] A. Moroni, R. Genesi, and D. Manstretta, "Analysis and design of a 54 GHz distributed 'hybrid' wave oscillator array with quadrature outputs," *IEEE J. Solid-State Circuits*, vol. 49, no. 5, pp. 1158–1172, May 2014.
- [6] M. B. Dayanik, N. Collins, and M. P. Flynn, "A 28.5–33.5 GHz fractional- N PLL using a 3rd order noise shaping time-to-digital converter with 176 fs resolution," in *Proc. 41st Eur. Solid-State Circuits Conf. (ESSCIRC)*, 2015, pp. 376–379.
- [7] D. Weyer, M. B. Dayanik, S. Jang, and M. P. Flynn, "A 36.3-to-38.2 GHz -216 dBc/Hz² 40 nm CMOS fractional- N FMCW chirp synthesizer PLL with a continuous-time bandpass delta-sigma time-to-digital converter," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2018, pp. 250–252.
- [8] M. Ferriss, B. Sadhu, A. Ryljakov, H. Ainspan, and D. Friedman, "A 13.1-to-28 GHz fractional- N PLL in 32 nm SOI CMOS with a $\Delta\Sigma$ noise-cancellation scheme," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2015, pp. 1–3.
- [9] L. Grimaldi et al., "A 30 GHz digital sub-sampling fractional- N PLL with 198 fs_{rms} jitter in 65 nm LP CMOS," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2019, pp. 268–270.
- [10] A. Hussein, S. Vasadi, M. Soliman, and J. Paramesh, "A 50-to-66 GHz 65 nm CMOS all-digital fractional- N PLL with 220 fs_{rms} jitter," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2017, pp. 326–327.
- [11] W. Wu, R. B. Staszewski, and J. R. Long, "A 56.4-to-63.4 GHz multi-rate all-digital fractional- N PLL for FMCW radar applications in 65 nm CMOS," *IEEE J. Solid-State Circuits*, vol. 49, no. 5, pp. 1081–1096, May 2014.
- [12] M. Hekmat, F. Aryanfar, J. Wei, V. Gadde, and R. Navid, "A 25 GHz fast-lock digital LC PLL with multiphase output using a magnetically-coupled loop of oscillators," *IEEE J. Solid-State Circuits*, vol. 50, no. 2, pp. 490–502, Feb. 2015.
- [13] W. Wu et al., "A 28-nm 75- fs_{rms} analog fractional- N sampling PLL with a highly linear DTC incorporating background DTC gain calibration and reference clock duty cycle correction," *IEEE J. Solid-State Circuits*, vol. 54, no. 5, pp. 1254–1265, May 2019.
- [14] H. Yoon et al., "A -31 dBc integrated-phase-noise 29 GHz fractional- N frequency synthesizer supporting multiple frequency bands for backward-compatible 5G using a frequency doubler and injection-locked frequency multipliers," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2018, pp. 366–368.
- [15] Y. Hu et al., "A 21.7-to-26.5 GHz charge-sharing locking quadrature PLL with implicit digital frequency-tracking loop achieving 75 fs jitter and -250 dB FoM ," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2020, pp. 276–278.
- [16] A. Li, S. Zheng, J. Yin, X. Luo, and H. C. Luong, "A 21–48 GHz subharmonic injection-locked fractional- N frequency synthesizer for multiband point-to-point backhaul communications," *IEEE J. Solid-State Circuits*, vol. 49, no. 8, pp. 1785–1799, Aug. 2014.

- [17] D. Park and S. Cho, "A 14.2 mW 2.55-to-3 GHz cascaded PLL with reference injection and 800 MHz delta-sigma modulator in 0.13 μm CMOS," *IEEE J. Solid-State Circuits*, vol. 47, no. 12, pp. 2989–2998, Dec. 2012.
- [18] L. Kong and B. Razavi, "A 2.4-GHz 6.4-mW fractional-N inductorless RF synthesizer," in *Proc. IEEE Symp. VLSI Circuits (VLSI-Circuits)*, Jun. 2016, pp. 1–2.
- [19] J.-H. Seol, D. Sylvester, D. Blaauw, and T. Jang, "A reference oversampling digital phase-locked loop with -240 dB FOM and -80 dBc reference spur," in *Proc. Symp. VLSI Circuits*, Jun. 2019, pp. 160–161.
- [20] J. Du et al., "A 24–31 GHz reference oversampling ADPLL achieving $\text{FoM}_{\text{jitter}-N}$ of -269.3 dB," in *Proc. Symp. VLSI Circuits*, Jun. 2021, pp. 1–2.
- [21] J. Qiu et al., "A 32 kHz-reference 2.4 GHz fractional-N oversampling PLL with 200 kHz loop bandwidth," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2021, pp. 454–456.
- [22] D. Yang et al., "A calibration-free triple-loop bang-bang PLL achieving 131 fs_{rms} jitter and -70 dBc fractional spurs," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2019, pp. 266–268.
- [23] M. Osada, Z. Xu, and T. Iizuka, "A 3.2-to-3.8 GHz calibration-free harmonic-mixer-based dual-feedback fractional-N PLL achieving -66 dBc worst-case in-band fractional spur," in *Proc. IEEE Symp. VLSI Circuits*, Jun. 2020, pp. 1–2.
- [24] D. Yang, D. Murphy, H. Darabi, A. Behzad, R. Ruby, and R. Parker, "An FBAR driven -261 dB FOM fractional-N PLL," in *Proc. IEEE Radio Freq. Integr. Circuits Symp. (RFIC)*, Jun. 2021, pp. 147–150.
- [25] D. Yang et al., "A sub-100 MHz reference-driven 25-to-28 GHz fractional-N PLL with -250 dB FoM," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2022, pp. 384–386.
- [26] H. Xu and A. A. Abidi, "Design methodology for phase-locked loops using binary (bang-bang) phase detectors," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 64, no. 7, pp. 1637–1650, Jul. 2017.
- [27] B. Razavi, "Jitter-power trade-offs in PLLs," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 68, no. 4, pp. 1381–1387, Apr. 2021.
- [28] X. Gao, E. A. M. Klumperink, P. F. J. Geraedts, and B. Nauta, "Jitter analysis and a benchmarking figure-of-merit for phase-locked loops," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 56, no. 2, pp. 117–121, Feb. 2009.
- [29] P. G. M. Baltus et al., "A 3.5-mW, 2.5-GHz diversity receiver and a 1.2-mW, 3.6-GHz VCO in silicon on anything," *IEEE J. Solid-State Circuits*, vol. 33, no. 12, pp. 2074–2079, Dec. 1998.
- [30] P. Kinget, *Integrated GHz Voltage Controlled Oscillators*. Boston, MA, USA: Springer, 1999, pp. 353–381, doi: [10.1007/978-1-4757-3047-0_17](https://doi.org/10.1007/978-1-4757-3047-0_17).
- [31] M. Z. Straayer and M. H. Perrott, "A multi-path gated ring oscillator TDC with first-order noise shaping," *IEEE J. Solid-State Circuits*, vol. 44, no. 4, pp. 1089–1098, Apr. 2008.
- [32] S. L. J. Gierkink, "Low-spur, low-phase-noise clock multiplier based on a combination of PLL and recirculating DLL with dual-pulse ring oscillator and self-correcting charge pump," *IEEE J. Solid-State Circuits*, vol. 43, no. 12, pp. 2967–2976, Dec. 2008.
- [33] S. Levantino, G. Marzin, C. Samori, and A. L. Lacaita, "A wideband fractional-N PLL with suppressed charge-pump noise and automatic loop filter calibration," *IEEE J. Solid-State Circuits*, vol. 48, no. 10, pp. 2419–2429, Oct. 2013.
- [34] M. Perrott, M. Trott, and C. Sodini, "A modeling approach for Σ - Δ fractional-N frequency synthesizers allowing straightforward noise analysis," *IEEE J. Solid-State Circuits*, vol. 37, no. 8, pp. 1028–1038, Aug. 2002.
- [35] D. Tasca, M. Zanuso, G. Marzin, S. Levantino, C. Samori, and A. L. Lacaita, "A 2.9–4.0-GHz fractional-N digital PLL with bang-bang phase detector and 560 fs_{rms} integrated jitter at 4.5-mW power," *IEEE J. Solid-State Circuits*, vol. 46, no. 12, pp. 2745–2758, Dec. 2011.
- [36] C.-M. Hsu, M. Z. Straayer, and M. H. Perrott, "A low-noise, wide-BW 3.6 GHz digital $\Delta\Sigma$ fractional-N frequency synthesizer with a noise-shaping time-to-digital converter and quantization noise cancellation," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2008, pp. 340–341.
- [37] S. Pamarti, L. Jansson, and I. Galton, "A wideband 2.4-GHz delta-sigma fractional-N PLL with 1-Mb/s in-loop modulation," *IEEE J. Solid-State Circuits*, vol. 39, no. 1, pp. 49–62, Jan. 2004.
- [38] R. E. Best, *Phase-Locked Loops Theory, Design, and Applications*, 2nd ed. New York, NY, USA: McGraw-Hill, 1993.
- [39] D. Scherer, B. S. Chan, F. H. Ives, W. J. J. Crilly, and W. D. Mathiesen, "Low-noise RF signal generator design," *Hewlett Packard J.*, vol. 32, no. 2, pp. 12–21, 1981.
- [40] S. Aniruddhan, S. Shekhar, and D. J. Allstot, "A CMOS 1.6 GHz dual-loop PLL with fourth-harmonic mixing," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 58, no. 5, pp. 860–867, May 2011.
- [41] K. J. Wang, A. Swaminathan, and I. Galton, "Spurious tone suppression techniques applied to a wide-bandwidth 2.4 GHz fractional-N PLL," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2008, pp. 342–343.
- [42] A. Mirzaei, M. E. Heidari, R. Bagheri, S. Chehrizi, and A. A. Abidi, "The quadrature LC oscillator: A complete portrait based on injection locking," *IEEE J. Solid-State Circuits*, vol. 42, no. 9, pp. 1916–1932, Sep. 2007.
- [43] A. Mirzaei and H. Darabi, "Mutual pulling between two oscillators," *IEEE J. Solid-State Circuits*, vol. 49, no. 2, pp. 360–372, Feb. 2014.
- [44] A. V. Oppenheim, A. S. Willsky, S. H. Nawab, G. M. Hernández, and I. T. Young, *Signals and Systems*, 1st ed. Englewood Cliffs, NJ, USA: Prentice-Hall, 1983.
- [45] X. Gao, E. A. M. Klumperink, M. Bohsali, and B. Nauta, "A low noise sub-sampling PLL in which divider noise is eliminated and PD/CP noise is not multiplied by N^2 ," *IEEE J. Solid-State Circuits*, vol. 44, no. 12, pp. 3253–3263, Dec. 2009.
- [46] K. Raczkowski, N. Markulic, B. Hershberg, and J. Craninckx, "A 9.2–12.7 GHz wideband fractional-N subsampling PLL in 28 nm CMOS with 280 fs_{rms} jitter," *IEEE J. Solid-State Circuits*, vol. 50, no. 5, pp. 1203–1213, May 2015.
- [47] W. Leng and A. A. Abidi, "Approximate equivalent circuits to understand tradeoffs in geometry of on-chip inductors," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 68, no. 3, pp. 975–988, Mar. 2021.
- [48] D. Murphy, J. J. Rael, and A. A. Abidi, "Phase noise in LC oscillators: A phasor-based analysis of a general result and of loaded Q ," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 57, no. 6, pp. 1187–1203, Jun. 2010.
- [49] D. Murphy and H. Darabi, "A 27-GHz quad-core CMOS oscillator with no mode ambiguity," *IEEE J. Solid-State Circuits*, vol. 53, no. 11, pp. 3208–3216, Nov. 2018.
- [50] A. Bhat and N. Krishnapura, "A 25-to-38 GHz, 195 dB FoMT LC QVCO in 65 nm LP CMOS using a 4-port dual-mode resonator for 5G radios," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2019, pp. 412–414.
- [51] L. Fanori, T. Mattsson, and P. Andreani, "A 2.4-to-5.3 GHz dual-core CMOS VCO with concentric 8-shaped coils," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2014, pp. 370–371.
- [52] W. Deng et al., "An 8.2-to-21.5 GHz dual-core quad-mode orthogonal-coupled VCO with concurrently dual-output using parallel 8-shaped resonator," in *Proc. IEEE Custom Integr. Circuits Conf. (CICC)*, Apr. 2021, pp. 1–2.
- [53] N. M. Neihart, D. J. Allstot, M. Miller, and P. Rakers, "Twisted inductors for low coupling mixed-signal and RF applications," in *Proc. IEEE Custom Integr. Circuits Conf.*, Sep. 2008, pp. 575–578.
- [54] E. Hegazi, H. Sjoland, and A. A. Abidi, "A filtering technique to lower LC oscillator phase noise," *IEEE J. Solid-State Circuits*, vol. 36, no. 12, pp. 1921–1930, Dec. 2001.
- [55] M. Garampazzi, P. M. Mendes, N. Codega, D. Manstretta, and R. Castello, "Analysis and design of a 195.6 dBc/Hz peak FoM P-N class-B oscillator with transformer-based tail filtering," *IEEE J. Solid-State Circuits*, vol. 50, no. 7, pp. 1657–1668, Jul. 2015.
- [56] L. Kong and B. Razavi, "A 2.4 GHz 4 mW integer-N inductorless RF synthesizer," *IEEE J. Solid-State Circuits*, vol. 51, no. 3, pp. 626–635, Mar. 2016.
- [57] D. Yang, "A multi-loop calibration-free phase-locked loop (PLL) for wideband clock generation," Ph.D. dissertation, Elect. Comp. Eng. Dept., Univ. California, Los Angeles, CA, USA, 2019.
- [58] T. Iizuka and A. A. Abidi, "FET-R-C circuits: A unified treatment—Part I: Signal transfer characteristics of a single-path," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 63, no. 9, pp. 1325–1336, Sep. 2016.
- [59] J. Kim et al., "A 76 fs_{rms} jitter and -40 dBc integrated-phase-noise 28-to-31 GHz frequency synthesizer based on digital sub-sampling PLL using optimally spaced voltage comparators and background loop-gain optimization," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2019, pp. 258–260.

- [60] J. Shin and H. Shin, "A 1.9–3.8 GHz $\Delta\Sigma$ fractional-N PLL frequency synthesizer with fast auto-calibration of loop bandwidth and VCO frequency," *IEEE J. Solid-State Circuits*, vol. 47, no. 3, pp. 665–675, Mar. 2012.
- [61] T.-H. Lin and Y.-J. Lai, "An agile VCO frequency calibration technique for a 10-GHz CMOS PLL," *IEEE J. Solid-State Circuits*, vol. 42, no. 2, pp. 340–349, Feb. 2007.
- [62] A. Aktas and M. Ismail, "CMOS PLL calibration techniques," *IEEE Circuits Devices Mag.*, vol. 20, no. 5, pp. 6–11, Sep./Oct. 2004.
- [63] R. Bagheri et al., "An 800-MHz–6-GHz software-defined wireless receiver in 90-nm CMOS," *IEEE J. Solid-State Circuits*, vol. 41, no. 12, pp. 2860–2876, Dec. 2006.
- [64] G. Li, L. Liu, Y. Tang, and E. Afshari, "A low-phase-noise wide-tuning-range oscillator based on resonant mode switching," *IEEE J. Solid-State Circuits*, vol. 47, no. 6, pp. 1295–1308, Jun. 2012.
- [65] Y. Wang, T. Zou, B. Chen, S. Ji, C. Zhang, and N. Yan, "A 7.9–14.3 GHz –243.3 dB FoMT sub-sampling PLL with transformer-based dual-mode VCO in 40 nm CMOS," in *Proc. IEEE Asian Solid-State Circuits Conf. (A-SSCC)*, Nov. 2021, pp. 1–3.



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